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First line investigations for clinically suspected myocarditis (discuss with on-call Cardiologist)

Diagnostic criteria for clinically suspected myocarditis (>1 symptom and >1 clinical criteria in the absence of pre-existing cardiovascular disease or extracardiac causes; if no symptoms then >2 clinical criteria) Caforio et al, Eur Heart J 2013

Symptoms:	Clinical criteria:
❖ Chest pain ❖ Recent onset dyspnoea on exercise or at rest ❖ Palpitations ❖ Syncope ❖ Aborted sudden cardiac death. ❖ Cardiogenic shock	❖ Newly abnormal 12 lead ECG: AV block, new BBB, ST/T changes, SVT incl AF, frequent PVCs, VT/VF, low QRS voltages, pathological Q waves ❖ Myocardiocytolysis markers: elevated Troponin (TnT or TnI) ❖ Functional and structural abnormalities on cardiac imaging (Echo/Angio/MRI): new regional wall motion abnormality or global systolic or diastolic dysfunction with or without ventricular dilatation, hypertrophy, pericardial effusion or intracavitary thrombi ❖ Tissue characterization by CMR: oedema and/or LGE of classical myocarditis pattern

Investigation	*Date(s)*	*Result*
CXR		Cardiothoracic ratio =
ECG		
24 hour ECG		

Sample	*Investigation*	*Date sent*	*Result*	*Lab*	*Sample*
Blood	FBC and film (incl. vacuolated lymphocytes)			Haem	EDTA
	ESR			Haem	EDTA - minimum 4 mls
	Iron/Transferrin/Ferritin			Haem	Plain clotted sample
	Serum Glucose			Biochem	Grey top
	UE, LFT, Bone, Mg CRP. Uric acid			Biochem	Lithium Heparin
	HS-TnI (High sensitivity Troponin I)			Biochem	Lithium Heparin
	Blood gas				Capillary tube
	Acute viral serology: CMV, EBV, Parvo B19 IgM, PCR CMV, EBV, Adenovirus PCR enterovirus EBNA if > 1 year old Parvo B19 IgG HIV antigen/antibody Based on clinical symptoms Mycoplasma, HSV, HBV, HCV testing under the respiratory testing			Virol	EDTA
	Bacterial culture			Micro	Blood culture bottle(s)

To be filed Patient Case notes
Updated 18/08/2025 by Dr Maria Ilina, Consultant Paediatric Cardiologist, Dr Conor Doherty, Consultant in Paediatric Infectious Disease RHC, Dr Celia Jackson, Consultant Virologist, GRI/QEUH, and Dr Alana Christie, Consultants in Paediatric Metabolic Medicine RHC

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	Borrelia (Lyme's disease) serology			Micro	EDTA
Blood	ASOT			Micro	Clotted blood
Throat swab	Bacterial culture (incl Group A strep)			Micro	Charcoal medium swab

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Sample	*Investigation*	*Date sent*	*Result*	*Lab*	*Sample*
Throat swab	Extended respiratory viral screen by PCR including: Influenza A&B RSV Adenovirus Mycoplasma Parainfluenza1&2 Human MPV Coronavirus Rhinovirus Enterovirus			Virol	Swab viral medium
Throat swab	HHV6-(HHV6 can reactivate or can be chromosomally integrated so a positive result may not indicate acute infection- DW virology if unsure)			Virol	Swab viral medium
Urine	Clean catch/catheter urine for MC&S			Micro	Universal container
Stool/rectal swab	Enterovirus PCR			Virol	Stool sample container

Check recent travel history.
If recent foreign travel, discuss rationale for 2nd line test with ID.

Check immunisation status.
MMR (infants under 1 – maternal rubella status)
DTP
Meningococcus
Pneumococcus

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First line investigations for dilated and hypertrophic cardiomyopathy (select tests in discussion with on-call Cardiologist)

Sample	***Investigation***	***Date sent***	***Result***	***Lab***	***Sample***
Blood	CK			Biochem	Lithium Heparin
	Amino Acids			Biochem	Lithium Heparin
	Acyl carnitine			Biochem	Lithium Heparin
	NT- pro BNP			Biochem	Call lab – labile sample!
	Thyroid Function			Biochem	Lithium Heparin
	Serum Vitamin D			Biochem	Lithium Heparin
	Parathyroid hormone (PTH)			Biochem	Lithium Heparin
Urine	Organic and amino acids			Biochem	Universal container
Urine	GAG and oligosaccharides			Biochem	Universal container
Blood	Ammonia			Biochem	Lithium Heparin
	Lactate, pyruvate (2 hours after carbohydrate – containing meal)			Biochem	Lithium Heparin
	DST (transferrin electrophoresis, age>3 weeks)			Biochem	Lithium Heparin
	Pompe Blood Spot Alpha Glucosidase			Biochem	Blood Spot Card
Children older than 3 years					
Blood	ASOT (if not already done as part of myocarditis screen)			Micro	Clotted blood
	Auto antibodies: ANA, ANCA, DNA Abs, rheumatoid factor, Anti-Ro. Anti-La.			Immun	Yellow IF

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2nd Line investigations for clinically suspected myocarditis and/or dilated cardiomyopathy (to be requested ONLY after discussion with the duty Paediatric ID and Cardiology Consultants)

Sample	*Investigation*	*Date sent*	*Result*	*Lab*	*Sample*
Discuss with Cardiology and/or Metabolic Consultant					
Blood	Copper			Biochem	Lithium Heparin
	Selenium			Biochem	Lithium Heparin
	Fasting lipid profile			Biochem	Lithium Heparin
	TTG-IgA (coeliac screen)			Biochem	Clotted sample
	Thiamine			Biochem	Lithium Heparin
If recent foreign travel or immunocompromised, discuss with Duty ID Consultant to choose from the following					
Blood	Toxoplasma serology			Virol	EDTA (ON DISCUSSION WITH ID)
Blood	Spirochaetal and rickettsial serology			Virol	EDTA (ON DISCUSSION WITH ID)
Blood	Special bacteriology culture for Brucella, Mycobacterium, Corynebacterium			Micro	Blood culture bottle (ON DISCUSSION WITH ID)
skin lesion swabs	PCR for VZV, HSV, other as directed by ID/Virology (if positive 1st line throat swab and/or skin lesions)			Virol	EDTA Swab (viral medium)
Stool	Protozoal infection (Trypanosoma, Entamoeba, Leishmania)			Micro	Stool sample container
Blood and Stool	Tests for parasitic infestation (trichinella, Echinococcus, Taenia)			Parasitology,	EDTA, Stool sample container
Throat and groin swab	Fungal cultures in immunocompromised			Micro	

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Other consultations (discuss with Cardiology)

Neurology consultation	
Metabolic consultation	

Genetic investigations (discuss with Cardiology and Genetics; parental consent and separate referral required; consider whole exome sequencing if diagnosis remains unclear)

Investigation	***Date sent***	***Sample/Lab information***
DCM gene panel		EDTA 1-3 ml (Molecular Genetics) – Genetic lab
HCM gene panel		EDTA 1-3 ml (Molecular Genetics) – Genetic lab
RAS-MAPKinase pathway		EDTA 1-3 ml (Molecular Genetics) – Genetic lab
Friedrich's Ataxia genetics		EDTA 1-3 ml (Molecular Genetics) – Genetic lab
LV non-compaction gene panel		EDTA 1-3 ml (Molecular Genetics) – Genetic lab
Barth syndrome (males) Cardiolipin analysis & TAZ gene sequencing		Cardiolipin sample 1- 3ml EDTA: Biochemistry – handwritten form http://www.uhbristol.nhs.uk/patients-and-visitors/your-hospitals/other-services-in-bristol/barthsyndromeservice/testing/ TAZ sequencing 1-3ml EDTA (separate sample to cardiolipin) send to genetic and they will forward to Bristol genetics laboratory, see link for details and request form. https://www.nbt.nhs.uk/sites/default/files/NHS%20Specialised%20Barth%20Syndrome%20Service.pdf

Optional investigations as directed by Cardiology, Metabolic and Neurology teams.

White cell and RBC enzymology
Fibroblast enzymology
Liver biopsy and enzymology
Marrow Aspiration
Skeletal muscle Biopsy
Cardiac muscle biopsy
DNA studies on lymphocytes or other tissue electrophoresis
Pompe disease screening
Serum ACE for Sarcoidosis